

Things You Wanted to Know!

Q1. Many circuits published in EFY that require a step-down centre-tap transformer having voltage ratings of 15V-0-15V, 18V-0-18V or 21V-0-21V, and current ratings of 1A, 1.5A, 2A, etc are not available in the market. Where can I get these transformers?

Anirvan
Kolkata

A1. This information is available in EFY's annual Electronics Industry Directory. You can also get this information online from our website www.eleb2b.com

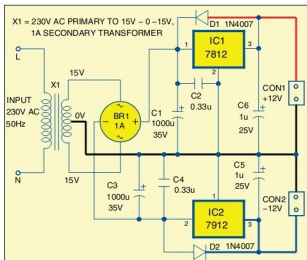
Q2. I have a couple of questions:
1. What are wearables? Please provide brief information about these.
2. What are AVR controllers?

Ashish Mulajkar
Pune (Maharashtra)

A2. **Wearables.** The terms wearables, wearable technology and wearable devices all refer to electronic technologies or sensors and computing devices that are incorporated in items of clothing and accessories, which can comfortably be worn on the body. To understand the subject, please read the Tech Focus articles in this issue. We have covered this subject in some previous issues as well.

AVR controllers. A microcontroller is a tiny computer on chip. It combines an arithmetic logic unit (ALU), memory (ROM, RAM or EEPROM), timers/counters, serial ports, input/output (I/O) ports, clock oscillators and a number of peripherals. These include small (micro) and control processes, objects and events. A number of companies are making controllers in billions; a large percentage of these find applications in embedded systems.

AVR microcontrollers are a product of Atmel. According to Atmel, an AVR is not an acronym and does not stand for anything in particular. However, it is



Circuit diagram of a dual power supply

commonly accepted that AVR stands for Alf and Vegard's RISC processor. Alf and Vegard are initials of two Norwegian students who conceived the architecture of AVR. AVR microcontrollers are divided into three groups:

- tinyAVR
- AVR (classic AVR)
- megaAVR

The difference between these devices mostly lies in available features. A tinyAVR microcontroller is a device with a lower pin-count or reduced feature set, as compared to the megaAVR. All AVR devices have the same basic instruction sets and memory organisation, so migrating from one device to another is usually easy. ATmega16 is a very popular AVR microcontroller. You will find many projects based on AVR microcontrollers in EFY.

Q3. I want to design a 12V dual power supply. Please help me choose the required components.

Suresh Jaganathan
Through e-mail

A3. The circuit diagram of a dual power supply is given in the figure above. Transformer X1 is a step-down transformer with a centre-tap. It steps down 230AC mains at its primary winding to 15V-0-15V AC voltage, 1A at its secondary. The stepped-down voltage is rectified by bridge rectifier BR1. The rectified DC voltage is smoothened out by electro-

lytic capacitors C1 and C3. Positive voltage from BR1 is input to positive regulator 7812 (IC1). Output of regulator IC1 is +12V, which is available at CON1.

Similarly, negative voltage from BR1 is input to negative regulator 7912 (IC2). Output of regulator IC2 is -12V, which is available at CON2. Capacitors C2, C4, C5 and C6 are de-couplers. These decouple ripple voltages spewed by voltage regulators on the power line. These should be soldered near the pins of regulators. Diodes D1 and D2 are for protection. The regulated voltage provides line as well as load regulation.

Q4. I am a hobbyist and would like to know about EPABX.

Susanta Payra
Kolkata

A4. A private branch exchange (PBX) is a telephone exchange that serves a particular business or office, as opposed to one that a common carrier or telephone company operates for many businesses or for the general public.

PBXes are also referred to as:

PABX - private automatic branch exchange, or

EPABX - electronic private automatic branch exchange

PBXes make connections among internal telephones of a private organisation, usually a business, and connect these to the public switched telephone network (PSTN) via trunk lines. PBXes are differentiated from key systems in that users of key systems manually select their outgoing lines, while PBXes select outgoing lines automatically.

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